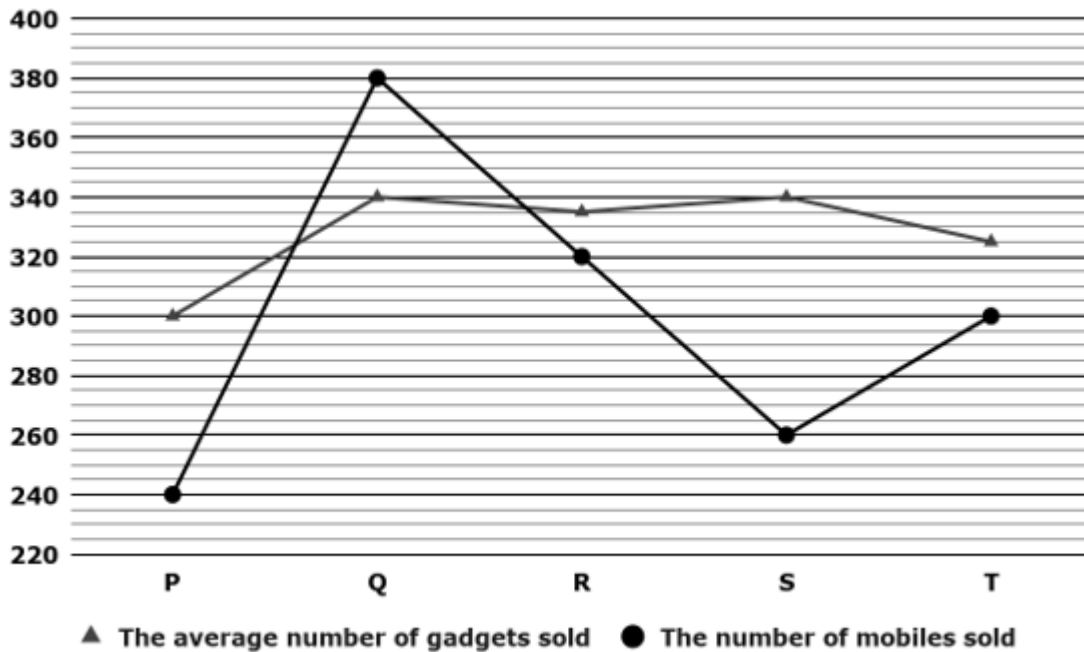


1. Questions

Study the following information carefully and answer the questions given below.

The line graph shows the average number of gadgets (laptops and mobiles) sold by five different companies namely P, Q, R, S and T along with the number of Mobiles sold by each company.

Note: The total number of gadgets sold = The number of laptops sold + The number of mobiles sold.



If the number of premium laptops sold by company S is 40% of the total laptops sold by it, then find the number of non-premium laptops sold by company S.

(Note: Company S sold only premium and non-premium laptops)

- 242
- 256
- 248
- 252
- 246

2. Questions

Find the ratio of the number of laptops sold by company P to the number of mobiles sold by company R.

- 7 : 6
- 9 : 8
- 8 : 7
- 9 : 5

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e. 5 : 4

3. Questions

The total number of mobiles sold by companies P and S together is what percentage of the total number of gadgets sold by company P?

- a. 87.5%
- b. 55.55%
- c. 83.33%
- d. 90%
- e. 62.5%

4. Questions

The total number of gadgets sold by company Q is how much more than that sold by company T?

- a. 50
- b. 60
- c. 20
- d. 40
- e. 30

5. Questions

If the difference between the number of laptops sold by companies R and Q is 'y', find the value of $(80 + y)$.

- a. 130
- b. 120
- c. 140
- d. 110
- e. 150

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6. Questions

Study the following information carefully and answer the questions given below.

A and B started a business, and A invested a certain amount for the entire year. B invested Rs. 4000 more than that of A, but withdrew his entire investment after 8 months. C joined the business 4 months after it started and invested Rs. 2000 less than B's investment. At the end of the year, the ratio of the profit share of A to C was 9 : 8.

If the total profit earned at the end of the year is Rs. 10260, find the profit share of A.

- a. Rs. 2260
- b. Rs. 3240
- c. Rs. 2180
- d. Rs. 3420
- e. Rs. 2340

7. Questions

The investment of C is what percentage more or less than the investment of A?

- a. 33.33% more
- b. 12.5% less
- c. 13.33% more
- d. 33.33% less
- e. 14.28% more

8. Questions

Find the ratio of the profit shares of A, B and C in the business at the end of the year.

- a. 11 : 9 : 7
- b. 10 : 8 : 5
- c. 7 : 10 : 8
- d. 8 : 7 : 9
- e. 9 : 10 : 8

9. Questions

The ratio of the speed of the stream to the upstream speed of a boat is 3 : 2. If the time taken by the boat to cover 20 km in still water is 2 hours, find the time taken by the boat to cover 112 km downstream.

- a. 9 hours
- b. 8 hours
- c. 3 hours
- d. 5 hours
- e. 7 hours

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10. Questions

Rs. x is invested at R% p.a simple interest for 15 years, it amounts to 4 times the principal. If Rs.

$(x + 400)$ is invested at $(R + 10)\%$ compounded annually for 2 years, it amounts to Rs. 5408, then find the value of x .

- a. 4800
- b. 3600
- c. 4200
- d. 2800
- e. 3200

11. Questions

P and Q together can complete a work in 12 days, while P alone can complete the same work in 36 days. P and Q start working together on it, but after 6 days, P leaves the work. In how many days can Q complete the remaining part of the work alone?

- a. 10 days
- b. 9 days
- c. 12 days
- d. 8 days
- e. 5 days

12. Questions

The average weight of three friends P, Q and R is 52 kg. If the average weight of P, Q and R is 20% less than the average weight of P, Q, R, S and T, find the weight of S and T together.

- a. 152 kg
- b. 138 kg
- c. 169 kg
- d. 145 kg
- e. 163 kg

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13. Questions

260 litres of mixture X containing milk and water in the ratio of 7 : 6, respectively, is mixed with another mixture Y containing milk and water in the ratio of 2:3, respectively. If the ratio of the quantity of milk to that of water in the resultant mixture is 1 : 1, then find the total quantity of mixture Y.

- a. 100 litres
- b. 120 litres
- c. 90 litres

d. 110 litres

e. 80 litres

14. Questions

If a number is increased by $y\%$ and then increased by 25% , the resultant number is 750, which is 250 less than the double of the original number. Find the value of 'y'.

a. 50

b. 10

c. 40

d. 30

e. 20

15. Questions

The ratio of the present ages of Rahul and Rohit is 5 : 3 respectively. 4 years ago, the sum of their ages was 48 years. If the average of the present ages of Rohit and Kiran is 26 years, then find the present age of Kiran.

a. 25 years

b. 37 years

c. 28 years

d. 31 years

e. 22 years

16. Questions

Aman bought two articles for Rs. 3200. He sold one article at a loss of 12% and the other at a profit of 12% . If the selling prices of both the articles are equal, then find the difference between their cost prices.

a. Rs. 426

b. Rs. 384

c. Rs. 484

d. Rs. 326

e. Rs. 452

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17. Questions

A train crosses a 450 meter-long bridge in 32 seconds, and the same train crosses a tunnel, which is 60% longer than the bridge, in 44 seconds. Find the length of the train.

- a. 180 m
- b. 240 m
- c. 270 m
- d. 300 m
- e. 250 m

18. Questions

The circumference of a circle is 176 cm. If the side of the square is 12 cm less than the radius of the circle, then find the area of the square.

- a. 256 cm^2
- b. 144 cm^2
- c. 81 cm^2
- d. 196 cm^2
- e. 484 cm^2

19. Questions

Sneha spends 10% of her income on rent, 25% on education, 35% on groceries, and she saves her remaining income. If the amount saved by Sneha is Rs. 1350, find the amount that Sneha spends on groceries.

- a. Rs. 1694
- b. Rs. 2195
- c. Rs. 1728
- d. Rs. 2174
- e. Rs. 1575

20. Questions

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A bag contains 4 red balls and x blue balls. If one ball is drawn at random from the bag, the probability of drawing a red ball is $\frac{4}{11}$. Find the total number of balls in the bag.

- a. 9
- b. 14
- c. 7
- d. 11
- e. 5

21. Questions

What value should come in the place of (?) in the following questions?

$$(4/9) * 441 + (5/8) * 384 = ? * 4$$

- a. 112
- b. 109
- c. 120
- d. 118
- e. 105

22. Questions

$$11(1/9)\% \text{ of } 720 + 37.5\% \text{ of } 160 = ? + 48$$

- a. 82
- b. 98
- c. 84
- d. 92
- e. 74

23. Questions

$$\sqrt{2025} * 6 + 14^3 \div 49 = ?$$

- a. 326
- b. 248
- c. 344
- d. 296
- e. 342

24. Questions

$$\sqrt[3]{512} + \sqrt{2704} - 13 * 4 = ?^3$$

- a. 3
- b. 4
- c. 2
- d. 5
- e. 6

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25. Questions

$$5\frac{1}{3} + 4\frac{1}{2} - 2\frac{5}{6} = ? - 1\frac{2}{3}$$

- a. $7(1/2)$
- b. $9(1/6)$
- c. $7(3/4)$
- d. $8(1/2)$
- e. $8(2/3)$

26. Questions

What approximate value will come in the place of (?) in the following questions?

$$(5399.97 \div 449.95) * \sqrt{168.89} + 14.23 = ?$$

- a. 150
- b. 140
- c. 120
- d. 170
- e. 130

27. Questions

$$? + 342.02 - 191.98 = 74.99 * 3.97$$

- a. 150
- b. 170
- c. 160
- d. 180
- e. 140

28. Questions

$$(8.97)^3 - (25.04 - \sqrt{26})^2 - (14.92)^2 = ?$$

- a. 125
- b. 110
- c. 114
- d. 120
- e. 104

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29. Questions

$$(344.91 \div 4.98) - (179.88 \div 8.94) = ?^2$$

- a. 9
- b. 5
- c. 7
- d. 3
- e. 6

30. Questions

$$139.92\% \text{ of } 499.89 + 24.94 * 2.12 = ?\% \text{ of } 749.95$$

- a. 120
- b. 100
- c. 80
- d. 110
- e. 90

31. Questions

What value will come in the place of (?) in the following number series?

24, 72, 36, 108, 54, ?

- a. 148
- b. 157
- c. 129
- d. 162
- e. 136

32. Questions

160, 123, 92, 63, ?, 21

- a. 40
- b. 42
- c. 30
- d. 37
- e. 45

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33. Questions**123, 131, ?, 204, 229, 445**

- a. 152
- b. 130
- c. 145
- d. 138
- e. 140

34. Questions**119, ?, 128, 101, 137, 92**

- a. 120
- b. 89
- c. 110
- d. 79
- e. 118

35. Questions**37, 50, 63, ?, 89, 102**

- a. 81
- b. 76
- c. 68
- d. 92
- e. 87

36. Questions**Find out the wrong number in the following number series.** @RankZen Banker**93, 95, 99, 105, 113, 120**

- a. 105
- b. 95
- c. 113
- d. 120
- e. 99

37. Questions**3, 6, 9, 17, 33, 65**

- a. 6
- b. 17
- c. 3
- d. 33
- e. 9

38. Questions**8, 13, 23, 40, 58, 83, 113**

- a. 13
- b. 23
- c. 113
- d. 8
- e. 40

39. Questions**16, 19, 24, 33, 48, 70, 104**

- a. 16
- b. 70
- c. 24
- d. 48
- e. 33

40. Questions**8640, 1440, 288, 72, 25, 12**

- a. 288
- b. 8640
- c. 25
- d. 1440
- e. 72

41. Questions

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The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 + 12x + 35 = 0$

II). $y^2 - 7y + 10 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be determined.
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 11x + 28 = 0$

II). $2y^2 - 9y + 10 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be determined.
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 - 24x + 143 = 0$

II). $y^2 - 23y + 132 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be determined.
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 = 196$

II). $y^2 - 37y + 322 = 0$

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- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be determined.
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^2 - 19x + 88 = 0$

II). $y^2 - 15y + 56 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be determined.
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

Company P:

The total number of gadgets sold by company P = $300 * 2 = 600$

The number of mobiles sold by company P = 240

The number of laptops sold by company P = $600 - 240 = 360$

Company Q:

The total number of gadgets sold by company Q = $340 * 2 = 680$

The number of mobiles sold by company Q = 380

The number of laptops sold by company Q = $680 - 380 = 300$

Company R:

The total number of gadgets sold by company R = $335 * 2 = 670$

The number of mobiles sold by company R = 320

The number of laptops sold by company R = $670 - 320 = 350$

Company S:

The total number of gadgets sold by company S = $340 * 2 = 680$

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The number of mobiles sold by company S = 260

The number of laptops sold by company S = $680 - 260 = 420$

Company T:

The total number of gadgets sold by company T = $325 * 2 = 650$

The number of mobiles sold by company T = 300

The number of laptops sold by company T = $650 - 300 = 350$

Company	The number of laptops sold	The number of mobiles sold	The total number of gadgets sold
P	360	240	600
Q	300	380	680
R	350	320	670
S	420	260	680
T	350	300	650
Total	1780	1500	3280

Answer: D

According to the question,

The number of premium laptops sold by company S = $420 * 40\% = 168$

The non-premium laptops sold by company S = $420 - 168 = 252$

Hence, the correct answer is option D.

2. Questions

Company P:

The total number of gadgets sold by company P = $300 * 2 = 600$ @RankZen Banker

The number of mobiles sold by company P = 240

The number of laptops sold by company P = $600 - 240 = 360$

Company Q:

The total number of gadgets sold by company Q = $340 * 2 = 680$

The number of mobiles sold by company Q = 380

The number of laptops sold by company Q = $680 - 380 = 300$

Company R:

The total number of gadgets sold by company R = $335 * 2 = 670$

The number of mobiles sold by company R = 320

The number of laptops sold by company R = $670 - 320 = 350$

Company S:

The total number of gadgets sold by company S = $340 * 2 = 680$

The number of mobiles sold by company S = 260

The number of laptops sold by company S = $680 - 260 = 420$

Company T:

The total number of gadgets sold by company T = $325 * 2 = 650$

The number of mobiles sold by company T = 300

The number of laptops sold by company T = $650 - 300 = 350$

Company	The number of laptops sold	The number of mobiles sold	The total number of gadgets sold
P	360	240	600
Q	300	380	680
R	350	320	670
S	420	260	680
T	350	300	650
Total	1780	1500	3280

Answer: B

The number of laptops sold by company P = 360

The number of mobiles sold by company R = 320

The required ratio = $360 : 320 = 9 : 8$

Hence, the correct answer is option B.

3. Questions

Company P:

The total number of gadgets sold by company P = $300 * 2 = 600$

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The number of mobiles sold by company P = 240

The number of laptops sold by company P = $600 - 240 = 360$

Company Q:

The total number of gadgets sold by company Q = $340 * 2 = 680$

The number of mobiles sold by company Q = 380

The number of laptops sold by company Q = $680 - 380 = 300$

Company R:

The total number of gadgets sold by company R = $335 * 2 = 670$

The number of mobiles sold by company R = 320

The number of laptops sold by company R = $670 - 320 = 350$

Company S:

The total number of gadgets sold by company S = $340 * 2 = 680$

The number of mobiles sold by company S = 260

The number of laptops sold by company S = $680 - 260 = 420$

Company T:

The total number of gadgets sold by company T = $325 * 2 = 650$

The number of mobiles sold by company T = 300

The number of laptops sold by company T = $650 - 300 = 350$

Company	The number of laptops sold	The number of mobiles sold	The total number of gadgets sold
P	360	240	600
Q	300	380	680
R	350	320	670
S	420	260	680
T	350	300	650
Total	1780	1500	3280

Answer: C

The total number of mobiles sold by companies P and S together = $240 + 260 = 500$

The total number of gadgets sold by company P = 600

The required percentage = $500 * 100/600 = 83.33\%$

Hence, the correct answer is option C.

4. Questions

Company P:

The total number of gadgets sold by company P = $300 * 2 = 600$

The number of mobiles sold by company P = 240

The number of laptops sold by company P = $600 - 240 = 360$

Company Q:

The total number of gadgets sold by company Q = $340 * 2 = 680$

The number of mobiles sold by company Q = 380

The number of laptops sold by company Q = $680 - 380 = 300$

Company R:

The total number of gadgets sold by company R = $335 * 2 = 670$

The number of mobiles sold by company R = 320

The number of laptops sold by company R = $670 - 320 = 350$

Company S:

The total number of gadgets sold by company S = $340 * 2 = 680$

The number of mobiles sold by company S = 260

The number of laptops sold by company S = $680 - 260 = 420$

Company T:

The total number of gadgets sold by company T = $325 * 2 = 650$

The number of mobiles sold by company T = 300

The number of laptops sold by company T = $650 - 300 = 350$

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Company	The number of laptops sold	The number of mobiles sold	The total number of gadgets sold
P	360	240	600
Q	300	380	680
R	350	320	670
S	420	260	680
T	350	300	650
Total	1780	1500	3280

Answer: E

The total number of gadgets sold by company Q = 680

The total number of gadgets sold by company T = 650

The required answer = $680 - 650 = 30$

Hence, the correct answer is option E.

5. Questions

Company P:

The total number of gadgets sold by company P = $300 * 2 = 600$

The number of mobiles sold by company P = 240

The number of laptops sold by company P = $600 - 240 = 360$

Company Q:

The total number of gadgets sold by company Q = $340 * 2 = 680$

The number of mobiles sold by company Q = 380

The number of laptops sold by company Q = $680 - 380 = 300$

Company R:

The total number of gadgets sold by company R = $335 * 2 = 670$

The number of mobiles sold by company R = 320

The number of laptops sold by company R = $670 - 320 = 350$

Company S:

The total number of gadgets sold by company S = $340 * 2 = 680$

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The number of mobiles sold by company S = 260

The number of laptops sold by company S = $680 - 260 = 420$

Company T:

The total number of gadgets sold by company T = $325 * 2 = 650$

The number of mobiles sold by company T = 300

The number of laptops sold by company T = $650 - 300 = 350$

Company	The number of laptops sold	The number of mobiles sold	The total number of gadgets sold
P	360	240	600
Q	300	380	680
R	350	320	670
S	420	260	680
T	350	300	650
Total	1780	1500	3280

Answer: A

The value of $y = 350 - 300 = 50$

The required value of $(80 + y) = 50 + 80 = 130$

Hence, the correct answer is option A.

6. Questions

Let the initial investment of A = x

The initial investment of B = $(x + 4000)$

The initial investment of C = $(x + 2000)$

The total investment of A = $x * 12 = 12x$

The total investment of B = $(x + 4000) * 8$

The total investment of C = $(x + 2000) * 8$

According to the question,

$$12x / (8x + 16000) = 9 / 8$$

$$4x = 3x + 6000$$

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$$x = 6000$$

The initial investment of A = Rs. 6000

The initial investment of B = Rs. (6000 + 4000) = Rs. 10000

The initial investment of C = Rs. (6000 + 2000) = Rs. 8000

The profit sharing ratio of A, B and C = $(6000 * 12) : (10000 * 8) : (8000 * 8) = 9 : 10 : 8$

Answer: D

The profit share of A = $10260 * 9/27 = \text{Rs. } 3420$

Hence, the correct answer is option D.

7. Questions

Let the initial investment of A = x

The initial investment of B = $(x + 4000)$

The initial investment of C = $(x + 2000)$

The total investment of A = $x * 12 = 12x$

The total investment of B = $(x + 4000) * 8$

The total investment of C = $(x + 2000) * 8$

According to the question,

$$12x / (8x + 16000) = 9 / 8$$

$$4x = 3x + 6000$$

$$x = 6000$$

The initial investment of A = Rs. 6000

The initial investment of B = Rs. (6000 + 4000) = Rs. 10000

The initial investment of C = Rs. (6000 + 2000) = Rs. 8000

The profit sharing ratio of A, B and C = $(6000 * 12) : (10000 * 8) : (8000 * 8) = 9 : 10 : 8$

Answer: A

The investment of C = Rs. 8000

The investment of A = Rs. 6000

The required percentage = $(8000 - 6000) * 100/6000 = 33.33\%$ more

Hence, the correct answer is option A.

8. Questions

Let the initial investment of A = x

The initial investment of B = $(x + 4000)$

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The initial investment of C = $(x + 2000)$

The total investment of A = $x * 12 = 12x$

The total investment of B = $(x + 4000) * 8$

The total investment of C = $(x + 2000) * 8$

According to the question,

$$12x / (8x + 16000) = 9 / 8$$

$$4x = 3x + 6000$$

$$x = 6000$$

The initial investment of A = Rs. 6000

The initial investment of B = Rs. $(6000 + 4000) = \text{Rs. } 10000$

The initial investment of C = Rs. $(6000 + 2000) = \text{Rs. } 8000$

The profit sharing ratio of A, B and C = $(6000 * 12) : (10000 * 8) : (8000 * 8) = 9 : 10 : 8$

Answer: E

The profit sharing ratio of A, B and C = $9 : 10 : 8$

Hence, the correct answer is option E.

9. Questions

Answer: E

Let the speed of the stream = $3a$ km/hr

The upstream speed of the boat = $2a$ km/hr

The speed of the boat in still water = $3a + 2a = 5a$ km/hr

According to the question,

$$20/5a = 2$$

$$a = 2$$

The speed of the boat in still water = $5 * 2 = 10$ km/hr

The speed of the stream = $3 * 2 = 6$ km/hr

The downstream speed of the boat = $10 + 6 = 16$ km/hr

The required time = $112/16 = 7$ hours

Hence, the correct answer is option E.

10. Questions

Answer: D

Formula used:

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The simple interest = $PRT/100$

The effective rate of compound interest = $a + b + ab/100$

The compound interest = $P [(1 + R/100)^n - 1]$

The compound amount = $P [(1 + R/100)^n]$

According to the question,

$$(4x - x) = x * 15 * R/100$$

$$3x = x * 15 * R/100$$

$$R = 20\%$$

The rate of compound interest = $20 + 10 = 30\%$

The effective rate of compound interest = $30 + 30 + (30 * 30/100) = 69\%$

So,

$$5408 = (x + 400) * 169/100$$

$$3200 = x + 400$$

$$x = 2800$$

Hence, the correct answer is option D.

11. Questions

Answer: B

P and Q together can complete a work in 12 days

P alone can complete the same work in 36 days

Let the total work = 36 units [LCM of 12 and 36]

The efficiency of P and Q together = $36/12 = 3$ units per day

The efficiency of P = $36/36 = 1$ unit per day

The efficiency of Q alone = $3 - 1 = 2$ units per day

The work done by P and Q in 6 days = $3 * 6 = 18$ units

The remaining work = $36 - 18 = 18$ units

The time taken by Q to complete the remaining work = $18/2 = 9$ days

Hence, the correct answer is option B.

12. Questions

Answer: C

Let the total weight of P, Q and R = $52 * 3 = 156$ kg

The average weight of all five friends = $52 * 100/80 = 65$ kg

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The total weight of all five friends = $65 * 5 = 325$ kg

The weight of S and T together = $325 - 156 = 169$ kg

Hence, the correct answer is option C.

13. Questions

Answer: A

The quantity of milk in mixture X = $260 * \frac{7}{13} = 140$ litres

The quantity of water in mixture X = $260 * \frac{6}{13} = 120$ litres

Let the quantity of milk in mixture Y = $2a$ litres

The quantity of water in mixture Y = $3a$ litres

According to the question,

$$140 + 2a = 120 + 3a$$

$$a = 20$$

The total quantity of mixture Y = $5a = 5 * 20 = 100$ litres

Hence, the correct answer is option A.

14. Questions

Answer: E

Let the original number = x

According to the question,

$$2x - 250 = 750$$

$$2x = 1000$$

$$x = 500$$

The original number = 500

So,

$$500 * \frac{(100 + y)}{100} * \frac{125}{100} = 750$$

$$100 + y = 120$$

$$y = 20$$

Hence, the correct answer is option E.

15. Questions

Answer: D

Let the present age of Rahul = $5a$ years

The present age of Rohit = $3a$ years

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The age of Rahul 4 years ago = $5a - 4$

The age of Rohit 4 years ago = $3a - 4$

According to the question,

$$(5a - 4) + (3a - 4) = 48$$

$$8a = 56$$

$$a = 7$$

The present age of Rohit = $3 * 7 = 21$ years

The total age of Rohit and Kiran = $26 * 2 = 52$ years

The present age of Kiran = $52 - 21 = 31$ years

Hence, the correct answer is option D.

16. Questions

Answer: B

Let the cost price of one article = Rs. P

The cost price of another article = Rs. $(3200 - P)$

According to the question,

$$P * 88/100 = (3200 - P) * 112/100$$

$$11P = (44800 - 14P)$$

$$25P = 44800$$

$$P = 1792$$

The cost price of one article = Rs. 1792

The cost price of another article = $(3200 - 1792) = \text{Rs. } 1408$

The required difference = $1792 - 1408 = \text{Rs. } 384$

Hence, the correct answer is option B.

17. Questions

Answer: C

Let the length of the train = T meters

The length of the bridge = 450 meters

The length of the tunnel = $450 * 160/100 = 720$ meters.

According to the question,

$$(T + 450)/32 = (T + 720)/44$$

$$(T + 450)/8 = (T + 720)/11$$

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$$11T + 4950 = 8T + 5760$$

$$3T = 810$$

$$T = 270$$

The length of the train = 270 meters.

Hence, the correct answer is option C.

18. Questions

Answer: A

Formula used:

The circumference of circle = $2\pi r$

The area of square = a^2

Let the radius of the circle = r

According to the question,

$$2 * (22/7) * r = 176$$

$$r = 28 \text{ cm}$$

$$\text{Side of the square} = 28 - 12 = 16 \text{ cm}$$

$$\text{The area of the square} = 16^2 = 256 \text{ cm}^2$$

Hence, the correct answer is option A.

19. Questions

Answer: E

Let the monthly income of Sneha = $100a$

$$\text{The total amount spent on all expenses} = 100a * (10 + 25 + 35)/100 = 70a$$

$$\text{The remaining savings} = 100a - 70a = 30a$$

According to the question,

$$30a = 1350$$

$$a = 45$$

$$\text{The total monthly income of Sneha} = 100 * 45 = \text{Rs. } 4500$$

$$\text{The amount spent on groceries} = 4500 * 35\% = \text{Rs. } 1575$$

Hence, the correct answer is option E.

20. Questions

Answer: D

Formula used:

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The probability = The favorable outcomes / The total outcomes

Let the total number of balls = $(4 + x)$

The number of red balls = 4

The favorable outcomes = ${}^4C_1 = 4$

The total outcomes = ${}^{(4+x)}C_1 = (4 + x)$

According to the question,

$$4 / (4 + x) = 4 / 11$$

$$4 + x = 11$$

$$x = 7$$

The total number of balls = $(4 + 7) = 11$

Hence, the correct answer is option D.

21. Questions

Answer: B

$$(4/9) * 441 + (5/8) * 384 = ? * 4$$

$$196 + 240 = ? * 4$$

$$436 = ? * 4$$

$$109 = ?$$

Hence, the correct answer is option B.

22. Questions

Answer: D

$$11(1/9)\% \text{ of } 720 + 37.5\% \text{ of } 160 = ? + 48$$

$$(1/9) * 720 + (3/8) * 160 = ? + 48$$

$$80 + 60 = ? + 48$$

$$140 = ? + 48$$

$$92 = ?$$

Hence, the correct answer is option D.

23. Questions

Answer: A

$$\sqrt{2025} * 6 + 14^3 \div 49 = ?$$

$$45 * 6 + 2744 \div 49 = ?$$

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$$270 + 56 = ?$$

$$326 = ?$$

Hence, the correct answer is option A.

24. Questions

Answer: C

$$\sqrt[3]{512} + \sqrt{2704} - 13 * 4 = ?^3$$

$$8 + 52 - 52 = ?^3$$

$$8 = ?^3$$

$$? = 2$$

Hence, the correct answer is option C.

25. Questions

Answer: E

$$5\frac{1}{3} + 4\frac{1}{2} - 2\frac{5}{6} = ? - 1\frac{2}{3}$$

$$(16/3) + (9/2) - (17/6) = ? - (5/3)$$

$$(32/6) + (27/6) - (17/6) = ? - (10/6)$$

$$(42/6) = ? - (10/6)$$

$$(52/6) = ?$$

$$8(2/3) = ?$$

Hence, the correct answer is option E.

26. Questions

Answer: D

$$(5399.97 \div 449.95) * \sqrt{168.89} + 14.23 = ?$$

$$(5400 \div 450) * \sqrt{169} + 14 = ?$$

$$12 * 13 + 14 = ?$$

$$156 + 14 = ?$$

$$170 = ?$$

Hence, the correct answer is option D.

27. Questions

Answer: A

$$? + 342.02 - 191.98 = 74.99 * 3.97$$

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$$? + 342 - 192 = 75 * 4$$

$$? + 150 = 300$$

$$? = 150$$

Hence, the correct answer is option A.

28. Questions

Answer: E

$$(8.97)^3 - (25.04 - \sqrt{26})^2 - (14.92)^2 = ?$$

$$(9)^3 - (25 - \sqrt{25})^2 - (15)^2 = ?$$

$$729 - (20)^2 - 225 = ?$$

$$729 - 400 - 225 = ?$$

$$104 = ?$$

Hence, the correct answer is option E.

29. Questions

Answer: C

$$(344.91 \div 4.98) - (179.88 \div 8.94) = ?^2$$

$$(345 \div 5) - (180 \div 9) = ?^2$$

$$69 - 20 = ?^2$$

$$49 = ?^2$$

$$? = 7$$

Hence, the correct answer is option C.

30. Questions

Answer: B

$$139.92\% \text{ of } 499.89 + 24.94 * 2.12 = ?\% \text{ of } 749.95$$

$$140\% \text{ of } 500 + 25 * 2 = ?\% \text{ of } 750$$

$$700 + 50 = ?\% \text{ of } 750$$

$$750 = ?\% \text{ of } 750$$

$$750 = ? * 7.5$$

$$? = 100$$

Hence, the correct answer is option B.

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31. Questions

Answer: D

$$24 * 3 = 72$$

$$72 \div 2 = 36$$

$$36 * 3 = 108$$

$$108 \div 2 = 54$$

$$54 * 3 = 162$$

Missing number: 162

Hence, the correct answer is option D.

32. Questions

Answer: A

$$160 - 37 = 123$$

$$123 - 31 = 92$$

$$92 - 29 = 63$$

$$63 - 23 = 40$$

$$40 - 19 = 21$$

Missing number: 40

Hence, the correct answer is option A.

33. Questions

Answer: E

$$123 + (2)^3 = 131$$

$$131 + (3)^2 = 140$$

$$140 + (4)^3 = 204$$

$$204 + (5)^2 = 229$$

$$229 + (6)^3 = 445$$

Missing number: 140

Hence, the correct answer is option E.

34. Questions

Answer: C

$$119 - 9 = 110$$

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$$110 + 18 = 128$$

$$128 - 27 = 101$$

$$101 + 36 = 137$$

$$137 - 45 = 92$$

Missing number: 110

Hence, the correct answer is option C.

35. Questions

Answer: B

$$37 + 13 = 50$$

$$50 + 13 = 63$$

$$63 + 13 = 76$$

$$76 + 13 = 89$$

$$89 + 13 = 102$$

Missing number: 76

Hence, the correct answer is option B.

36. Questions

Answer: D

$$93 + 2 = 95$$

$$95 + 4 = 99$$

$$99 + 6 = 105$$

$$105 + 8 = 113$$

$$113 + 10 = 123$$

Wrong number: 120

Right number: 123

Hence, the correct answer is option D.

37. Questions

Answer: A

$$3 * 2 - 1 = 5$$

$$5 * 2 - 1 = 9$$

$$9 * 2 - 1 = 17$$

$$17 * 2 - 1 = 33$$

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$$33 * 2 - 1 = 65$$

Wrong number: 6

Right number: 5

Hence, the correct answer is option A.

38. Questions

Answer: E

$$8 + 5 = 13$$

$$13 + 10 = 23$$

$$23 + 15 = 38$$

$$38 + 20 = 58$$

$$58 + 25 = 83$$

$$83 + 30 = 113$$

Wrong number: 40

Right number: 38

Hence, the correct answer is option E.

39. Questions

Answer: B

$$16 \quad 19 \quad 24 \quad 33 \quad 48 \quad 71 \quad 104$$

$$+3 \quad +5 \quad +9 \quad +15 \quad +23 \quad +33$$

$$+2 \quad +4 \quad +6 \quad +8 \quad +10$$

Wrong number: 70

Right number: 71

Hence, the correct answer is option B.

40. Questions

Answer: C

$$8640 \div 6 = 1440$$

$$1440 \div 5 = 288$$

$$288 \div 4 = 72$$

$$72 \div 3 = 24$$

$$24 \div 2 = 12$$

Wrong number: 25

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Right number: 24

Hence, the correct answer is option C.

41. Questions

Answer: D

I). $x^2 + 12x + 35 = 0$

$$x^2 + 7x + 5x + 35 = 0$$

$$x(x + 7) + 5(x + 7) = 0$$

$$(x + 5)(x + 7) = 0$$

$$x = -5 \text{ or } -7$$

II). $y^2 - 7y + 10 = 0$

$$y^2 - 5y - 2y + 10 = 0$$

$$y(y - 5) - 2(y - 5) = 0$$

$$(y - 2)(y - 5) = 0$$

$$y = 2 \text{ or } 5$$

So, $x < y$

Hence, the correct answer is option D.

42. Questions

Answer: A

I). $x^2 - 11x + 28 = 0$

$$x^2 - 7x - 4x + 28 = 0$$

$$x(x - 7) - 4(x - 7) = 0$$

$$(x - 4)(x - 7) = 0$$

$$x = 4 \text{ or } 7$$

II). $2y^2 - 9y + 10 = 0$

$$2y^2 - 5y - 4y + 10 = 0$$

$$y(2y - 5) - 2(2y - 5) = 0$$

$$(y - 2)(2y - 5) = 0$$

$$y = 2 \text{ or } 5/2$$

So, $x > y$

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Hence, the correct answer is option A.

43. Questions

Answer: C

I). $x^2 - 24x + 143 = 0$

$$x^2 - 13x - 11x + 143 = 0$$

$$x(x - 13) - 11(x - 13) = 0$$

$$(x - 11)(x - 13) = 0$$

$$x = 11 \text{ or } 13$$

II). $y^2 - 23y + 132 = 0$

$$y^2 - 12y - 11y + 132 = 0$$

$$y(y - 12) - 11(y - 12) = 0$$

$$(y - 11)(y - 12) = 0$$

$$y = 11 \text{ or } 12$$

So, $x = y$ or relationship cannot be determined.

Hence, the correct answer is option C.

44. Questions

Answer: E

I). $x^2 = 196$

$$x = 14 \text{ or } -14$$

II). $y^2 - 37y + 322 = 0$

$$y^2 - 23y - 14y + 322 = 0$$

$$y(y - 23) - 14(y - 23) = 0$$

$$(y - 14)(y - 23) = 0$$

$$y = 14 \text{ or } 23$$

$$\text{So, } x \leq y$$

Hence, the correct answer is option E.

45. Questions

Answer: B

I). $x^2 - 19x + 88 = 0$

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$$x^2 - 11x - 8x + 88 = 0$$

$$x(x - 11) - 8(x - 11) = 0$$

$$(x - 8)(x - 11) = 0$$

$$x = 8 \text{ or } 11$$

II). $y^2 - 15y + 56 = 0$

$$y^2 - 8y - 7y + 56 = 0$$

$$y(y - 8) - 7(y - 8) = 0$$

$$(y - 8)(y - 7) = 0$$

$$y = 8 \text{ or } 7$$

So, $x \geq y$

Hence, the correct answer is option B.

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